Quiz 3 [25 min]

Check *ONLY* T if the given statement in the question is correct or *ONLY* F if not.

Correct Answer 4pts, No Answer -1 pts, Wrong Answer -2 pts. If the total points < 0, your score will be set to 0.

1. 10 cards must be minimally selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen. **(F)**

2. For integer n > 1, $\sum_{k=0}^{n}(-1)^k\binom{n}{k} = 0$ **(T)**

3. $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$, where n, k are positive integers with $n \geq k$. **(T)**

4. The equation $x_1 + x_2 + x_3 = 11$ have 76 solutions if x_1, x_2, and x_3 are nonnegative integers. **(F)**

5. We proved the statement "There is no largest prime number" as the following: Assume there is a largest prime number p. Then for $r = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot ... \cdot p$, $r + 1$ is a prime larger than p. Now, let P be the assertion that there is no largest prime and Q be the assertion that p is the largest prime. Let's consider the formal structure of the proof. We assume ¬P is true. Then, ¬P → Q is true and Q → ¬ Q. We obtain ¬P → ¬ Q. Then Q is true and ¬Q is true is inferred by applying modus tollens and modus ponens. By conjunction ¬Q ∧ Q is true. Therefore, the theorem is true. **(F)**

6. Suppose that n is in Z^+. $f(n) = (n^n + n \cdot 2^n + 5^n)(n! + 100^n)$. Big-O estimate for $f(n)$ is $O(n^n \cdot 100^n)$. **(F)**

7. The following procedure X sorts correctly given n real numbers $a_1, a_2, \ldots, a_n$ with $n \geq 2$:

procedure $X(a_1, a_2, \ldots, a_n$: real numbers with $n \geq 2)$

for $j := 2$ **to** n
  $i := 1$
  while $a_j > a_i$
    $i := i + 1$
  $m := a_j$
  for $k := 0$ **to** $j - i - 1$
    $a_{j-k} := a_{j-k-1}$
  $a_i := m$

$\{a_1, \ldots, a_n$ is in increasing order$\}$ **(T)**

8. Showing that there is a unique element x such that P(x) is the same as proving the statement

$$\exists x(P(x) \land \forall y\,(y \neq x \rightarrow P(y))).$$ **(F)**

9. The following proof for $\overline{A \cap B} = \overline{A} \cup \overline{B}$ is correct. **(T)**

$$
\begin{aligned}
\overline{A \cap B} &= \{x \mid x \in \overline{A \cap B}\} \\
&= \{x \mid x \notin A \cap B\} \\
&= \{x \mid \neg(x \in A \cap B)\} \\
&= \{x \mid \neg(x \in A \wedge x \in B)\} \\
&= \{x \mid \neg(x \in A) \vee \neg(x \in B)\} \\
&= \{x \mid x \notin A \vee x \notin B\} \\
&= \{x \mid x \in \overline{A} \vee x \in \overline{B}\} \\
&= \{x \mid x \in \overline{A} \cup \overline{B}\} = \overline{A} \cup \overline{B}
\end{aligned}
$$

10. The following proof is correct for showing that $(q \wedge (p \rightarrow \neg q)) \rightarrow \neg p$ is a tautology using propositional equivalence and the laws of logic. **(T)**

$$
\begin{aligned}
& (q \wedge (p \rightarrow \neg q)) \rightarrow \neg p \\
\Longleftrightarrow\ & (q \wedge (\neg p \vee \neg q)) \rightarrow \neg p \\
\Longleftrightarrow\ & ((q \wedge \neg p) \vee (q \wedge \neg q)) \rightarrow \neg p \\
\Longleftrightarrow\ & (q \wedge \neg p) \rightarrow \neg p \\
\Longleftrightarrow\ & \neg(q \wedge \neg p) \vee \neg p \\
\Longleftrightarrow\ & (\neg q \vee p) \vee \neg p \\
\Longleftrightarrow\ & \neg q \vee (p \vee \neg p)
\end{aligned}
$$

11. The following proof is correct for showing that $f(x)$ and $g(x)$ are functions from the set of real numbers to the set of real numbers. If $f(x)$ is $O(g(x))$ then $g(x)$ is $\Omega(f(x))$. $f(x)$ is $O(g(x))$ means that there are constants C and k s.t. $|f(x)| \le C|g(x)|$ for all $x > k$. We can get $\frac{1}{C}|f(x)| \le |g(x)|$. Take $C' = \frac{1}{C}$, then $C'|f(x)| \le |g(x)|$. Therefore $g(x)$ is $\Omega(f(x))$. **(T)**

12. The following proof is correct for disproving that if $f(x)$ is $O(g(x))$, then $2^{f(x)}$ is $2^{O(g(x))}$. Let $f(x) = 2x$ and $g(x) = x$, then $f(x)$ is $O(g(x))$. Now $2^{f(x)} = 2^{2x} = 4^x$, and $2^{g(x)} = 2^x$, and 4^x is not $O(2^x)$. **(T)**